# B3.2 Transport (HL)

What the Cell · DP Biology · worksheet · HL statements B3.2.11–16

**Name** _____ **Date** _____ **Total** _____ / 40

**About 40 minutes,** or one section at a time. The number of marks shows how many separate points to make.

## Section A: Recall

8 marks

**1** **Define** tissue fluid. B3.2.11 **[1]**

**2** **State** one component of plasma that is almost absent from tissue fluid. B3.2.12 **[1]**

**3** **Name** the valve between the left atrium and the left ventricle. B3.2.15 **[1]**

**4** **State** where the sinoatrial node is found. B3.2.16 **[1]**

**5** **Outline** two features of lymph vessels. B3.2.13 **[2]**

**6** **State** two differences between the heart of a bony fish and the heart of a mammal. B3.2.14 **[2]**

## Section B: Understanding

16 marks

**7** **Explain** how tissue fluid is formed and reabsorbed along a capillary. B3.2.11 **[3]**

**8** **Explain** the advantage of a double circulation over a single circulation. B3.2.14 **[3]**

**9** **Explain** why the wall of the left ventricle is thicker than the wall of the right ventricle. B3.2.15 **[2]**

**10** **Explain** how substances are exchanged between tissue fluid and the cells of a tissue. B3.2.12 **[3]**

**11** **Explain** how cardiac muscle is adapted to the work of the heart. B3.2.15 **[3]**

**12** **Explain** the importance of the delay at the atrioventricular node. B3.2.16 **[2]**

## Section C: Working with evidence

10 marks

All data in this section are invented for practice. The patterns are realistic.

**13** B3.2.16

Pressures were recorded in the left side of the heart of a person at rest during one cardiac cycle of 0.8 s.

| **Time / s** | **0** | **0.1** | **0.2** | **0.3** | **0.4** | **0.5** | **0.6** |
|---|---|---|---|---|---|---|---|
| **Left atrium / mmHg** | 6 | 9 | 5 | 7 | 10 | 5 | 6 |
| **Left ventricle / mmHg** | 5 | 8 | 70 | 118 | 90 | 4 | 5 |
| **Aorta / mmHg** | 80 | 77 | 75 | 116 | 100 | 90 | 85 |

a **State** the systolic and diastolic pressures in the aorta. **[2]**

b **Calculate** the pulse pressure. **[1]**

c **Identify** one time when the atrioventricular valve is open. **[1]**

d **Identify** the time at which both valves are shut. **[1]**

e **Deduce**, with a reason, whether the semilunar valve is open at 0.3 s. **[2]**

f **Calculate** the heart rate. **[1]**

g **Explain** why the pressure in the aorta stays high while the ventricle is relaxed. **[2]**

## Section D: Longer answer

6 marks

**14** **Describe** the events in the left side of the heart during one cardiac cycle, starting with the firing of the sinoatrial node. B3.2.16 **[6]**

What the Cell. Original questions written for this course.